



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 1 REPORT

Internship Program — Batch 2026

Submitted By:	Ahmar Hayat
Department:	Web Dev
Submitted To:	Team lead
Date:	June 2026

1. Introduction

This report presents the work completed for Task 9 of the M-Tech Production & Marketing Web Development Internship Program, Batch 2026.

M-Tech Production & Marketing, based in Haripur, KPK, builds data-driven websites. Until now we only worked with static data. This task is part of Phase 2: Frontend Development (Core) and covers Modern JavaScript: Asynchronous JS, Promises, Async/Await, and Fetch API to get data from the internet.

2. Objectives

The main objectives of this task were:

1. To understand Synchronous vs Asynchronous JavaScript.
2. To learn Promises and how to handle them with `.then().catch()`.
3. To learn Async/Await syntax for cleaner async code.
4. To learn Fetch API to get data from external APIs.
5. To learn ES6 Modules to organize code.

3. Work Done

1. Watched Video: "Async JavaScript" by Web Dev Simplified.
2. Watched Video: "JavaScript Promises" by The Net Ninja.
3. Watched Video: "Fetch API Crash Course" by Traversy Media.
4. Read MDN Docs for Async/Await and Fetch API.
5. Built "Weather Widget App" using Fetch API + OpenWeatherMap API.
6. Used `async/await` to get data and display it on the page.

4. Challenges Faced

1. Callback Hell: At first I tried nested `.then()` and it got messy. Async/Await fixed it.
2. API Key: OpenWeatherMap needs an API key. I had to sign up for free.

5. Key Learnings

1. Synchronous vs Asynchronous:
Synchronous = code runs line by line.
Asynchronous = code can run in background. APIs are async.
2. Promises: A Promise is a "promise" that we will get data later.

```
fetch(url)
  .then(response => response.json())
  .then(data => console.log(data))
  .catch(error => console.log(error))
```
3. Async/Await: Cleaner way to write Promises. Looks like synchronous code.

```
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const getData = async () => {
  try {
    const response = await fetch(url);
    const data = await response.json();
    console.log(data);
  } catch (error) {
    console.log(error);
  }
}
```

Fetch API: Built-in browser function to make HTTP requests. No library needed.

4. Used to GET data from APIs like Weather, News, Products.
5. ES6 Modules: Split code into files.

```
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// utils.js
export const formatTemp = (temp) => `${temp}°C`
// main.js
import { formatTemp } from './utils.js'
```

6. Conclusion

Before this, my websites had fake data. Now I can connect to any API and show live data. This is exactly what we will do in Phase 5: Weather Widget API.

The biggest win was async/await. It makes code 10x easier to read than .then() chains. try...catch for error handling is also critical.

Project: Weather Widget App

M-Tech Weather App

Weather in Haripur

31°C 

Cloudy and warm